

Reduced Recurrent-State Persistence Reproduces Selected Psilocybin-Related Working-Memory Signatures in Trained Recurrent Networks

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Acute psilocybin does not produce a uniform loss of cognitive performance: human studies instead report response slowing, greater impairment at higher working-memory load, sensitivity to longer intervals, and vulnerability to competing information. We asked whether any of seven abstract perturbations of task-trained recurrent neural networks (RNNs) reproduced this selective behavioural pattern. Five independently trained fixation-gated circular-memory networks and ten independently trained N-back networks were evaluated on 1,024 trials or sequences per condition. We tested synaptic-drive gain, heterogeneous drive gain, sensory-input gain, distractor-window gain, recurrent gain, carried-state persistence, and a conserved effective time-constant change. A carried-state persistence scale of 0.95 was the only setting to reproduce the complete descriptive pattern. It increased post-cue settling by 0.194 ± 0.252 model steps while changing ordinary delay-20 angular error by only $1.38 \pm 6.17\%$; long-minus-short delay selectivity was 0.264 ± 0.140 , distractor-minus-clean selectivity was 0.159 ± 0.080 , and 2-back-minus-0-back selectivity was 0.250 ± 0.077 (mean $\pm$ SD across trained seeds). Every contrast had the predicted direction in all five circular or all ten N-back networks. However, the slowing was small and the unperturbed distractor condition was weak. Reduced state persistence is therefore computationally sufficient to produce several selected cross-task signatures, but the result does not identify a biological action of psilocybin or establish mechanistic uniqueness.

INTRODUCTION

Working memory requires a neural system to preserve task-relevant information after its eliciting input has disappeared. Recurrent excitation, attractor dynamics, and slow manifolds provide candidate mechanisms through which an internal state can remain stable while still supporting a later response [1, 2]. A perturbation of these dynamics need not destroy memory immediately. A weak change may instead expose selective vulnerability when information must be maintained for longer, protected from irrelevant input, or updated under greater task load.

The acute human behavioural literature motivates precisely such a selective target. Psilocybin impaired 2-back performance more clearly than 0-back performance in a letter N-back task [3]. It also impaired multiple-object tracking while spatial working-memory span was comparatively spared in a small within-subject study [4]. Longer interval reproduction was more affected than shorter intervals, with spatial span impairment emerging at the higher dose [5]. In a memory-guided delayed-response task, reaction time increased without a clear reduction in correct responses [6]. A recent meta-analysis similarly found a more robust acute effect on reaction time than accuracy across attention and executive tasks, while cautioning that the evidence was heterogeneous and at risk of bias [7]. Together, these findings define qualitative dissociations rather than a single target level of performance.

Candidate network manipulations were motivated by, but not equated with, psychedelic mechanisms. REBUS relates relaxed high-level precision to altered post-synaptic gain and afferent sensitivity [8]; a receptor-

informed whole-brain model operationalised 5-HT_{2A} effects through spatially heterogeneous response gain [9]. Neither framework specifies a unique perturbation of a trained working-memory RNN. We therefore treated gain, recurrent weighting, sensory sensitivity, state persistence, and effective timescale as competing computational interventions.

We evaluated these interventions in two complementary task families. The fixation-gated circular task measured response settling, delay dependence, and distractor sensitivity. The N-back task measured selective impairment at higher updating load. We hypothesised that a credible candidate would reproduce several dissociations across independently trained networks rather than merely increase average error. This study is a candidate-versus-native-baseline analysis; generic-noise matching and biological identification were outside its scope.

METHODS

Fixation-gated circular delayed-response task

Each circular-task trial comprised pre-cue fixation, a circular cue, a silent memory delay, and a response period (Fig. 1A). Stimulus angle $\theta \in [0, 2\pi)$ was encoded over 32 evenly spaced preferred angles μ_j as

$$c_j(\theta) = \exp\{\kappa[\cos(\theta - \mu_j) - 1]\}, \quad \kappa = 8. \quad (1)$$

The fixation input and output remained high until the response cue. Circular outputs were deliberately silent before response and then reported the remembered angle. Pre-cue and cue durations varied during training;

evaluation used clean delays of 10, 20, 40, and 80 model steps. A separate delay-20 condition inserted a five-step irrelevant circular cue midway through the delay. The networks had not been trained with this distractor, making it an out-of-distribution robustness probe rather than a second trained task.

Context-cued N-back task

The N-back task presented 20 sequential items drawn from six stimulus identities (Fig. 1B). Two context channels instructed either 0-back or 2-back behaviour. In 0-back, the network classified whether the current item matched a fixed target identity. In 2-back, it classified whether the current item matched the item presented two positions previously. Sequences contained six matches; the 2-back generator additionally controlled one-back lures. The first two items were excluded from scored outcomes. The output consisted of match and non-match logits.

Recurrent networks and checkpoint replication

Both task families used 64-unit $\tanh$ recurrent networks with $dt = 20$, $\tau = 100$, and $\alpha = dt/\tau = 0.2$. The native update was

$$z_t = W_{\text{in}}x_t + b_{\text{in}} + W_{\text{rec}}h_{t-1} + b_{\text{rec}}, \quad (2)$$

$$h_t = \tanh[(1 - \alpha)h_{t-1} + \alpha z_t]. \quad (3)$$

The circular model had 33 inputs and outputs: 32 tuned channels plus fixation. The N-back model had six stimulus inputs, two rule-context inputs, and two outputs. Perturbations were applied only at evaluation; no model was retrained. Five competent circular checkpoints (seeds 20260714–20260718) and ten competent N-back checkpoints (seeds 20260912–20260921) were retained. Independently trained checkpoint, not trial or technical replicate, was the replication unit.

Candidate perturbations

We evaluated seven single-operator families (Table I). For synaptic-drive gain, the weight-derived input and recurrent drive was multiplied by scalar g , while learned biases remained outside the gain. Fixed heterogeneous gain replaced g with a positive unit-wise vector $\mathbf{g}$ drawn from a log-normal distribution and normalised to population mean one; three fixed vectors were treated as technical replicates and averaged within checkpoint. Sensory-input gain scaled only stimulus-channel input drive, leaving rule and fixation channels unchanged. Distractor-window

gain applied this operation only during the irrelevant circular cue. Recurrent gain scaled $W_{\text{rec}}h_{t-1}$ while leaving recurrent bias unchanged.

State persistence changed the carried-state term but retained the native drive coefficient:

$$h_t = \tanh[p(1 - \alpha)h_{t-1} + \alpha z_t], \quad (4)$$

where p is the persistence scale. This is an asymmetric intervention and is not identical to response gain. The conserved time-constant operator instead set $\alpha' = \min(\alpha/s, 1)$ and used

$$h_t = \tanh[(1 - \alpha')h_{t-1} + \alpha' z_t], \quad (5)$$

so carried-state and drive coefficients continued to sum to one.

Evaluation and behavioural signature

Each circular checkpoint, operator, strength, condition, and delay used 1,024 trials. Each N-back checkpoint, operator, strength, and load used 1,024 20-item sequences. Native and perturbed evaluations reused identical generated task samples within checkpoint and cell.

Circular response accuracy was mean absolute angular error. Post-cue settling was the first response step at which angular error was below 15° and population-vector amplitude exceeded 50% of the native reference amplitude. Trials that did not settle were capped at the 25-step response duration, yielding restricted mean settling time. Settling was interpreted only when fixation accuracy was at least 0.90 and at least 80% of trials settled. ‘‘Slowing with preservation’’ required a positive settling change at clean delay 20 while proportional angular-error impairment remained at or below 0.20.

For circular condition c , proportional impairment was

$$I_c = \frac{E_{\text{perturbed},c} - E_{\text{native},c}}{E_{\text{native},c}}, \quad (6)$$

where E is mean angular error. Delay selectivity was $I_{\text{delay } 80} - I_{\text{delay } 10}$; distractor selectivity was $I_{\text{distractor } 20} - I_{\text{clean } 20}$.

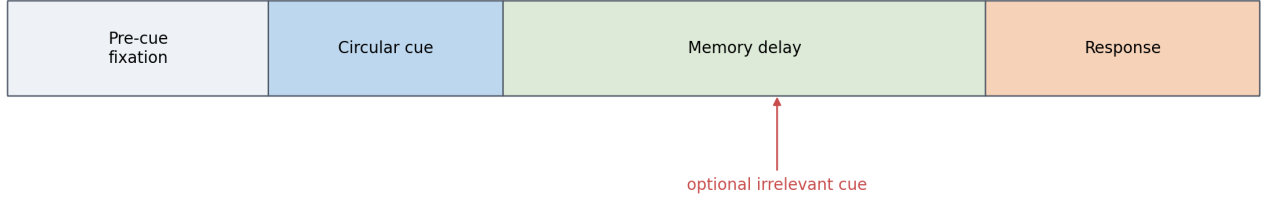
N-back discriminability was $D = H - F$, where H and F are hit and false-alarm rates. Condition-normalised impairment was

$$J_c = \frac{D_{\text{native},c} - D_{\text{perturbed},c}}{D_{\text{native},c}}, \quad (7)$$

and load selectivity was $J_{2\text{back}} - J_{0\text{back}}$. Positive values indicate disproportionate high-load impairment.

The analysis was descriptive. For each operator-strength setting we reported the mean contrast and fraction of checkpoint seeds in the predicted direction. A complete pattern required positive mean slowing with

A Fixation-gated circular delayed-response task



Clean delays: 10, 20, 40 or 80 steps; the remembered angle is reported only after the go cue.

B Context-cued 0-back and 2-back task

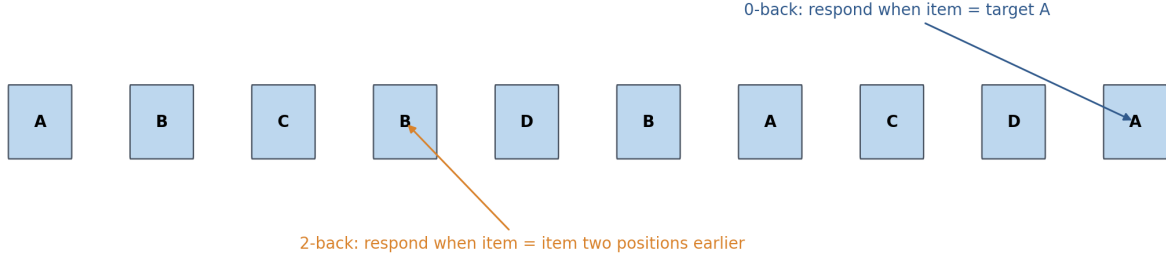


FIG. 1. **Complementary working-memory tasks.** **A**, The fixation-gated circular task required a continuous cue angle to be maintained across variable clean delays and reported after the go cue; an optional irrelevant delay-period cue tested distractor vulnerability. **B**, Context inputs switched the same N-back network between a fixed-target 0-back rule and a 2-back updating rule. The tasks supplied settling, delay, distractor, and updating-load contrasts rather than a single performance level.

TABLE I. **Candidate perturbations and fixed evaluation grids.** Neutral settings reproduce the native forward pass. Distractor-window gain was defined only for the circular distractor condition; all other operators were evaluated in both task families.

ID	Perturbation	Computational intervention	Strengths
P1	Synaptic-drive gain	Scalar gain on input and recurrent weight drive	0.90, 0.95, 1.00, 1.05, 1.10
P2	Heterogeneous drive gain	Fixed mean-one unit-wise log-normal gain	log-SD 0.00, 0.10, 0.20, 0.30
P3a	Sensory-input gain	Gain on stimulus-channel input contribution	0.80, 0.90, 1.00, 1.10, 1.20
P3b	Distractor-window gain	P3a restricted to irrelevant cue window	1.00, 1.10, 1.25, 1.50
P4	Recurrent gain	Scalar gain on recurrent weight contribution	0.90, 0.95, 1.00, 1.05, 1.10
P6	State persistence	Scale p on carried-state coefficient	0.90, 0.95, 1.00, 1.05, 1.10
P7	Effective time constant	Conserved update using $\alpha' = \alpha/s$	0.80, 0.90, 1.00, 1.10, 1.25

preservation, distractor selectivity, and N-back load selectivity, with a majority of checkpoint seeds agreeing; delay selectivity supplied additional supporting evidence. No participant-level inference, confirmatory p values, or multiple-comparison claims were made.

RESULTS

The full screen isolated one complete cross-task profile

The perturbations were not interchangeable. Across 29 shared operator-strength profiles, only state persistence $p = 0.95$ reproduced the complete descriptive pattern

(Fig. 2). It showed slowing with preservation, long-delay selectivity, and distractor selectivity in all five circular checkpoints, together with load selectivity in all ten N-back checkpoints. Neutral settings returned zero contrast. Other perturbations often reproduced one or two components, but not the full combination.

A 5% persistence reduction preserved ordinary accuracy while selectively affecting demanding conditions

At $p = 0.95$, clean delay-20 angular error changed by $1.38 \pm 6.17\%$ across circular checkpoints, satisfying the preservation criterion in all five seeds (Table II). Re-

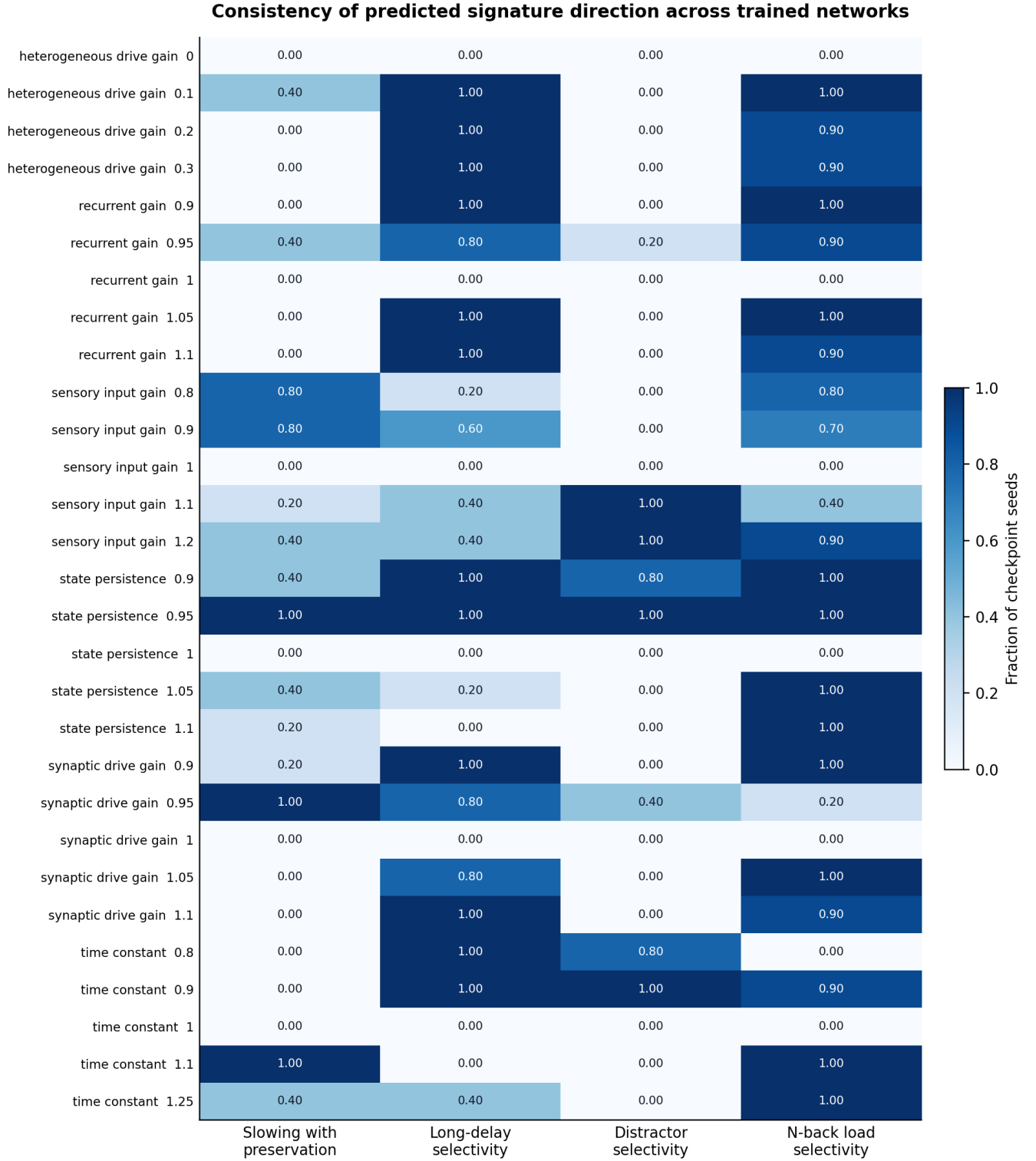


FIG. 2. **Cross-task signature screen.** Rows are every operator-strength setting evaluated in both task families; columns show the fraction of independent checkpoint seeds with the predicted signature direction. Circular fractions have denominator five and N-back fractions denominator ten. State persistence 0.95 was the only row with complete agreement across all four features. Values quantify directional consistency, not participant-level probability or statistical significance.

stricted mean settling increased by 0.194 ± 0.252 steps. Every seed showed a positive change, but the range was only 0.010–0.561 steps; the result is therefore a small, replicated response-dynamics analogue rather than a large reaction-time effect.

Long-minus-short delay selectivity was 0.264 ± 0.140 and positive in 5/5 checkpoints. Distractor-minus-clean selectivity was 0.159 ± 0.080 and also positive in 5/5. In the N-back networks, 2-back-minus-0-back selectivity was 0.250 ± 0.077 , positive in 10/10 checkpoints, and ranged from 0.110 to 0.363. Thus the leading perturbation did not merely increase average error: it preferentially affected longer maintenance, competing input, and higher updating load.

The persistence effect was dose-like but not monotonic across all features

Reducing persistence further to 0.90 enlarged mean settling, delay, and N-back load effects (Fig. 3). However, slowing with preservation passed only 2/5 circular checkpoints, so the stronger intervention no longer expressed the selective accuracy-preserving pattern. Increasing persistence above one continued to impair N-back preferentially but reversed the circular delay and distractor contrasts. The joint signature was therefore specific to a weak reduction around 0.95 rather than monotonic damage at any non-neutral strength.

Alternative operators reproduced partial signatures

Sensory-input gain 1.20 produced distractor selectivity in 5/5 circular checkpoints and load selectivity in 9/10 N-back checkpoints, but mean settling became slightly faster and delay selectivity was inconsistent. Effective time-constant scale 0.90 produced delay and distractor selectivity but also accelerated settling. Recurrent gain 0.95 showed delay selectivity in 4/5 and load selectivity in 9/10 checkpoints, but distractor selectivity in only 1/5. Heterogeneous drive gain frequently produced delay and N-back effects without distractor selectivity. These near-misses constrain the result: matching one difficult condition was insufficient to qualify as the complete pattern.

TABLE II. **State persistence 0.95.** Values are mean $\pm$ SD across five circular or ten N-back checkpoint seeds.

Outcome	Contrast	Consistency
Delay-20 error	0.014 ± 0.062	preserved 5/5
Settling change	0.194 ± 0.252 steps	5/5 positive
Delay selectivity	0.264 ± 0.140	5/5 positive
Distractor selectivity	0.159 ± 0.080	5/5 positive
N-back load selectivity	0.250 ± 0.077	10/10 positive

The distractor contrast remained validity-limited

All 945 circular result rows passed the fixation gate, all ten N-back native checkpoints passed competence, and all $p = 0.95$ clean delay-20 cells were latency-valid. However, the unperturbed distractor condition was already weak: mean angular error ranged from 25.8° to 50.4° , and only 26.7–35.7% of native distractor trials settled. Increasing the trial count reduced sampling uncertainty but could not repair this out-of-distribution baseline. The distractor direction is reproducible, but its magnitude may reflect floor or headroom effects and cannot support a strong Carter-style recovery claim in isolation.

DISCUSSION

The full candidate screen identified reduced carried-state persistence as the strongest computational account of the selected cross-task behavioural pattern. Scaling the carried-state coefficient by 0.95 left ordinary circular accuracy approximately intact while modestly slowing response settling, increasing vulnerability at longer delays and under distraction, and selectively impairing 2-back discriminability. The result replicated across all five circular and all ten N-back checkpoints.

Equation 4 suggests a coherent model-level explanation. Each update retains slightly less of the preceding hidden state without simultaneously rescaling incoming drive. Information is therefore not erased at once; rather, its stability is weakened cumulatively. Short and simple conditions remain relatively intact, whereas longer maintenance, competing input, and repeated updating expose the loss. This interpretation is supported by the delay and load contrasts, although the present behavioural analyses do not yet directly demonstrate a changed attractor or slow manifold.

Importantly, “state persistence” is an operational property of this implemented recurrence, not a measured neurobiological parameter. The result shows computational sufficiency: this intervention can generate the selected qualitative pattern. It does not show that psilocybin reduces neural persistence, that 5-HT2A activation maps to $p = 0.95$, or that the mechanism is unique. A generic-damage comparator was deliberately postponed, and several alternative operators reproduced individual components. Subsequent analyses should therefore examine hidden-state drift and stability and, separately, test specificity against non-neutral matched-cost controls.

Two limitations particularly constrain the present inference. First, post-cue settling is not literal reaction time because response onset was externally imposed; moreover, the mean change was less than one model step. Second, the distractor condition lacked a competent native baseline. A weaker irrelevant cue or a separately trained distractor-capable task is required before the

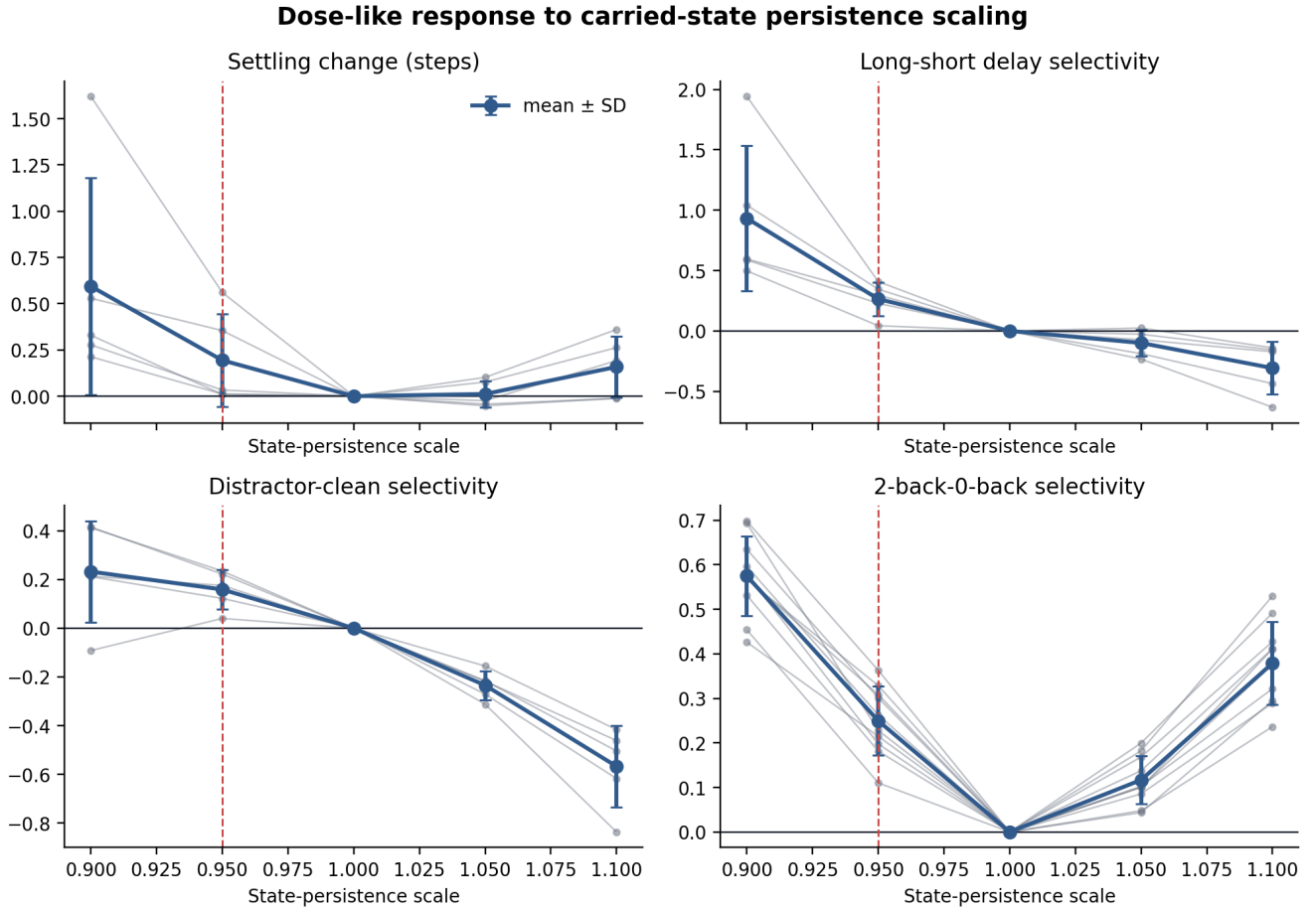


FIG. 3. **Dose-like response to carried-state persistence scaling.** Grey lines are independently trained checkpoint seeds; blue points and error bars are mean $\pm$ SD. The dashed red line marks the leading scale 0.95 and the horizontal line marks no change from native performance. A stronger reduction increased several impairments but also violated relative preservation, whereas increased persistence reversed circular delay and distractor selectivity.

Carter-style dissociation can be treated as secure. The descriptive grid also contains multiple operator-strength profiles but no confirmatory inferential family. Checkpoint consistency demonstrates robustness across trained models, not population inference about human participants.

The immediate next experiment should densify persistence scaling around 0.95, quantify delay-period representational drift, and repeat the clean, delay, and N-back contrasts on independently banked task samples. The distractor amplitude should first be calibrated to retain usable native performance. A later specificity analysis can compare the retained candidate with generic perturbation at comparable clean-task cost without changing the candidate-versus-baseline result reported here.

CONCLUSION

A 5% reduction in carried-state persistence was the only tested RNN perturbation to reproduce the complete pre-specified descriptive pattern: relative preservation with slower settling, greater long-delay and distractor costs, and selective 2-back impairment. The consistency across independently trained networks makes reduced persistence a credible candidate for further mechanistic analysis. The result nevertheless remains an abstract computational sufficiency finding, with small settling effects and a validity-limited distractor condition, rather than evidence for a biological mechanism of psilocybin.

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